



The role of behavioral inflexibility and impulsivity in explosive outbursts among youth with Tourette syndrome

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Abstract

Several children living with Tourette syndrome (TS) also present with recurrent episodes of sudden and intense anger, which are known as explosive outbursts. Nevertheless, individual characteristics influencing the onset of such outbursts remain poorly understood. In adults, impulsivity has been found to be a risk factor for explosive outbursts, but there are no such data among children. Additionally, behavioral inflexibility, which is conceptualized as a pattern of rigid, inflexible behavior in situations where it would be beneficial to adapt one's behavior to external demands, constitutes a trigger for anger outbursts in other neurodevelopmental conditions. However, its role in TS has not been studied yet. Therefore, the aim of this study was to assess whether impulsivity and behavioral inflexibility were associated with explosive outbursts. Three hundred and seventeen (317) parents of children (aged 6–14, 74.8% male) living with TS completed a series of online questionnaires. Parents completed the Behavioral Inflexibility Scale (BIS), the short-version of the UPPS-P Impulsivity Inventory, and the Rage Attack Questionnaire-Revised (RAQ-R) to assess behavioral inflexibility, impulsivity, and explosive outbursts, respectively. Parents also reported on socio-demographic and clinical characteristics of their child. Parent-rated impulsivity and behavioral inflexibility were significantly associated with explosive outbursts, as assessed with the RAQ-R. However, their interaction was not a significant predictor of explosive outbursts. Exploratory analyses also identified that several clinical characteristics, including tic severity and specific co-occurring conditions such as internalizing disorders, learning disorders, and sleep problems, were associated with explosive outbursts. Our results revealed that behavioral inflexibility, impulsivity, as well as several clinical characteristics were risk factors for explosive outbursts. Explosive outbursts are highly detrimental to the individual and familial functioning. Therefore, identifying risk factors for such outbursts is important given that it may lead to new targets for interventions aiming to decrease the frequency and intensity of explosive outbursts.

Keywords Tourette syndrome · Tics · Explosive outbursts · Rage attacks · Behavioral inflexibility · Impulsivity

Introduction

Tourette syndrome (TS) is a neurodevelopmental disorder characterized by the presence of multiple motor tics and at least one phonic tic [2]. In addition to tics, up to 90% of children living with TS also have other neurodevelopmental, emotional, or behavioral difficulties, such as attention deficit hyperactivity disorder (ADHD), depression, anxiety, oppositional defiant disorder, for instance [16, 32]. Another feature that several children with TS must deal with are explosive outbursts, which are also known as rage attacks [55].

Explosive outbursts are sudden episodes of intense anger in which the response is markedly disproportionate to the trigger [3, 13, 19]. Those outbursts are often defined as

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developmentally inappropriate [19]. They can take the form of verbal and/or physical aggression towards other individuals or objects [3]. Following outbursts, children may be calm or tired, and often express remorse or guilt [13]. Recent data suggest that approximately 50% of children with TS have frequent explosive outbursts [55]. When present, explosive outbursts lead to important disruptions in family functioning [22, 60].

While explosive outbursts are common among children with TS and are a leading cause of familial disruption, the exact causes of those outbursts remain understudied [19]. Reduced functional brain connectivity among regions involved in action selection and planning, emotional regulation, and impulse control has been reported among adults with TS and explosive outbursts [4]. Also, the presence of common co-occurring conditions, such as ADHD [17, 43, 46, 49, 55], obsessive-compulsive phenomena [17, 35, 46, 61], or autism spectrum disorder [50] has been associated with an increased risk of presenting with explosive outbursts.

Transdiagnostic traits associated with the aforementioned conditions may constitute precipitating factors leading to explosive outbursts. For instance, impulsivity, a common trait among individuals with ADHD, has been associated with explosive outbursts among adults with TS [46]. Additionally, co-occurring ADHD among individuals with TS is associated with increased inhibitory difficulties [44]. Also, a few factors have been identified as potential triggers of explosive outbursts, such as being reprimanded or criticized, not being able to do what was planned, having plans or routines changed, or being told to stop a game or activity [3, 12, 13, 40]. Such triggers may result from behavioral inflexibility, a transdiagnostic feature present in autism [8, 39] and obsessive-compulsive disorder (OCD) [26]. Behavioral inflexibility refers to a pattern of rigid, inflexible behavior in situations where it would be beneficial to adapt one's behavior to external demands [39]. To this date, behavioral inflexibility has not been formally assessed in TS. However, in autism [57] and Fragile X syndrome [34], behavioral inflexibility has been associated with anger outbursts. Behavioral inflexibility is also closely related to the concept of insistence on sameness [39], a common feature of autism spectrum disorder [7] which has been associated with increased irritability [42].

Previous work has shown an association between behavioral rigidity and impulsivity [14, 45]. In other populations, it is reported that the combination of elevated impulsivity and behavioral rigidity can be associated with a number of adverse effects, such as problematic internet use [37, 41], high body mass index [21, 23], suicide attempts [52], as well

as depressive and anxiety symptoms [65]. Impulsivity may thus potentiate the adverse effects of inflexibility (and vice versa), which could have negative consequences in children at risk for explosive outbursts. However, the combination of impulsivity and rigid behaviors and thoughts has yet to be investigated in TS.

The current study aimed to fill a few gaps the scientific literature. While an association between impulsivity and explosive outbursts has been demonstrated in adults, it is not known if such association exists in youth. Furthermore, the potentials effects of behavioral inflexibility on explosive outbursts have not been formally assessed in TS. The combination of high impulsivity and more rigid behavior may also put children with TS at increased risk for explosive outbursts, but this remains poorly understood. Therefore, the first aim of this study was to assess the association between behavioral inflexibility and explosive outbursts. The second aim was to assess the association between impulsivity and explosive episodes. For these first two aims, we expected that behavioral inflexibility and impulsivity would be both positively associated with explosive outbursts. The third aim was to assess the moderating role of impulsivity on the relationship between behavioral inflexibility and explosive episodes. We hypothesized that in children with high levels of impulsivity, the relationship between behavioral inflexibility and explosive episodes would be stronger than in children with low levels of impulsivity.

Methods

Participants

Participants were parents of children diagnosed with TS. They were mostly recruited online, through posts on social media, as well as ads and flyers shared in events targeting parents of children with TS. Inclusion criteria consisted of: (i) having a child aged 6–14 years old who has received a diagnosis of TS; (ii) living with that child at least 50% of time; (iii) being able to understand French or English; and (iv) completing the questionnaires allowing to assess at least one of the study objectives. Child medication intake or the presence of co-occurring conditions were not ground for exclusion. This study was approved by the human research ethics board (Comité d'éthique de la recherche avec des êtres humains) of the Université du Québec à Montréal (approval date: 2023-07-11; approval number: 2024-5786) and was conducted according to the principles of the Declaration of Helsinki. Participants provided consent prior to completing any study measure.

Measures

Participants first completed a sociodemographic and clinical questionnaire, where they provided information about their own and their child's demographics, as well as their child's clinical history and symptoms.

The Rage Attack Questionnaire-Revised [RAQ-R; 46] was used to assess the severity of explosive outbursts among participants' children. The RAQ-R is a 22-item inventory assessing several aspects of explosive outbursts. Items are rated from 0 to 3, for a possible maximum score of 66 (larger score suggesting increased severity). The RAQ-R has shown excellent internal consistency ($\alpha=0.93$ to 0.97) [46, 48] as well as good convergent validity with constructs such as aggression and hostility ($r = 0.68$) and low frustration tolerance ($r = 0.69$) [48].

Behavioral inflexibility was assessed with the Behavioral Inflexibility Scale [BIS; 39], which has 38 items rated on a 6-point scale (0 to 5). Higher scores on the BIS indicate enhanced behavioral inflexibility. The BIS has excellent psychometric properties (internal consistency: $\alpha=0.97$; test-retest reliability: $r = 0.92$) [39].

Impulsivity was assessed with the short version of the UPPS-P impulsivity scale [6]. The UPPS-P scale assesses impulsivity along 5 dimensions: negative urgency, positive urgency, lack of premeditation, lack of perseverance, and risk tasking. Negative and positive urgency respectively refer to the tendency to act impulsively when experiencing negative and positive emotions. The UPPS-S has 20 items rated on a scale from 1 to 4, for a total score ranging from 20 to 80. Larger scores indicate more impulsivity. The UPPS-P has good internal consistency ($\alpha=0.70$ to 0.84) and test-retest reliability ($r = 0.84$ to 0.92) [6].

Tic severity was assessed with the Parent Tic Questionnaire [PTQ; 15], a questionnaire assessing the frequency and intensity of the 14 most common motor and phonic tics. The PTQ has good internal consistency ($\alpha=0.70$ to 0.84) and test-retest reliability (ICC= 0.84 to 0.89), as well as good convergent validity with the Yale Global Tic Severity Scale ($r=0.53$ to 0.68) [53].

Procedures

Interested participants clicked on a hyperlink or scanned the QR code on the study flyer, which led them to a Qualtrics survey. Participants were able to complete the survey in French or English. The first part of the Qualtrics survey consisted of the consent form, which had to be completed prior to accessing the questionnaires. This study was part of a larger project investigating the developmental course of explosive outbursts as well as their impact on family functioning [28]. Therefore, in addition to the aforementioned

study measures, participants were also asked to complete the Family Assessment Device [24], the Parenting Stress Index-Short Form [1], and the Social Support Questionnaire-Short Form [56]. The order of questionnaires in the survey is provided in the supplement. Findings pertaining to those measures will be reported elsewhere. The median completion time for the total questionnaire was 42.4 min. Data were collected between January 24, 2024, and October 21, 2025.

Among participants meeting inclusion/exclusion criteria, we removed survey entries from robots or that appeared as fake responses, from participants who responded twice¹, and those with a duration of less than 15 min. If participants did not complete the survey but dropped out after completing the BIS, they were included in the study as long as their total response time was proportional to the number of items completed, as determined by an expected time-per-item ratio calculated using a rule-of-three approach. Three random attention checks were included in the survey. Participants who failed at least half of the attention checks were not included in statistical analyses. Any questionnaire that included more than 10% of items being answered with "I prefer not to answer" was not included in statistical analyses. Participants whose responses did not allow testing at least one objective (i.e., no valid responses in RAQ-R, BIS, or UPPS-P) were excluded from the current study.

Statistical analyses

Statistical analyses were performed with SPSS (version 29). To assess whether explosive outbursts were respectively associated with behavioral inflexibility and impulsivity, we conducted two correlations predicting RAQ-R score with the BIS and UPPS-P, respectively. The five subscales of the UPPS-P were entered as predictors in a multiple regression analysis to identify if one or more subscales predicted explosive outbursts over and above the other subscales. The PROCESS macro [version 3.5; 30] was used to assess the moderating role of the UPPS-P on the association between BIS and RAQ-R. We also performed exploratory t-tests to investigate whether co-occurring conditions had an impact on BIS and UPPS-P scores. Exploratory correlations were performed to assess the relationship between RAQ-R scores and tic severity, number of co-occurring diagnoses, and age. Finally, we performed exploratory t-tests to assess the impact of co-occurring conditions and sex on explosive outbursts.

¹ If a participant responded twice, we kept the response with the larger completion percentage. If a participant completed the full survey twice or had the same percentage of completion at multiple occurrences, we kept the initial response.

Results

Among the 733 entries in the Qualtrics survey, 550 participants met the inclusion criteria. After removal of invalid entries, we were left with 317 participants who provided enough data in order to answer at least one of the study objectives (i.e., they at least completed the socio-demographic questionnaire, the RAQ-R, and the BIS). The full survey was completed by 249 participants. The participants' flow diagram is provided in Figure S1. Participants' geographic distribution is provided in Figure S2. Demographics from participants (parents) included in the current study are presented in Table 1. Demographic and clinical data from children are provided in Table 2. Average scores on clinical scales are presented in Table 3.

With regards to our first aim, we found a significant and positive association between the BIS and the RAQ-R [$r(315)=0.50$, $p < 0.001$] (Fig. 1A). Exploratory analyses were performed to test the associations between BIS scores and co-occurring diagnoses (Table S1). Shortly, higher BIS

Table 1 Demographic characteristics of study participants (parents; $n=317$)

Age in years, mean (SD)	40.6 (5.7)
Sex, number of males (%)	22 (6.9%)
Gender, number (%)	
Men	15 (94.6)
Women	300 (4.7)
Non-binary	2 (0.6)
Ethnicity, number (%)	
Caucasian/European	288 (90.9%)
Native/First Nation/Aboriginal	3 (0.9%)
South Asian	1 (0.3%)
East Asian	3 (0.3%)
Latin American	6 (1.9%)
African	1 (0.3%)
Other	11 (3.5%)
Prefer not to answer	4 (1.3%)
Country of residence, number (%)	
Canada	162 (51.1%)
France	66 (20.8%)
United Kingdom	50 (15.8%)
United States of America	24 (7.6%)
Other ¹	15 (4.7%)
Family income, number (%)	
Less than \$25,000	11 (3.5%)
Between \$25,000 and \$50,000	33 (10.4%)
Between \$50,001 and \$75,000	73 (23.0%)
Between \$75,001 and \$100,000	58 (18.3%)
More than \$100,001	121 (38.2%)
Prefer not to answer	21 (6.6%)

Note: Income is reported in Canadian dollars (CAD). *ADHD* attention deficit hyperactivity disorder, *OCD* obsessive-compulsive disorder, *SD* standard deviation.¹ Other countries include Australia, Belgium, Brazil, Congo, Ireland, Poland, Spain, Sweden, Venezuela

Table 2 Demographic and clinical characteristics of children

Age in years, mean (SD)	11.6 (2.2)
Sex, number of males (%)	237 (74.8%)
Gender, number (%)	
Boys	242 (76.3)
Girls	72 (22.7)
Non-binary	2 (0.6)
Other	1 (0.9)
Children diagnosed by, number (%)	
Psychologist/Neuropsychologist/Psychotherapist	58 (18.3%)
Neurologist	118 (37.2%)
General practitioner	31 (8.8%)
Psychiatrist/Child psychiatrist	74 (23.3%)
Pediatrician	25 (7.9%)
Other/Unspecified/Prefer not to answer	11 (3.5%)
Medication status, number (%)	
On any psychotropic medication	195 (61.5%)
Stimulants	50 (15.8%)
α -Agonists	99 (31.2%)
Neuroleptics	88 (27.8%)
SSRIs	28 (8.8%)
Atomoxetine	12 (3.8%)
Antiseizure medications	10 (3.2%)
Other	10 (3.2%)
Concomitant diagnoses, number (%)	
ADHD	221 (69.7%)
OCD	89 (28.1%)
Internalizing disorders	194 (61.2%)
Developmental coordination disorder	40 (12.6%)
Learning disorders	77 (24.3%)
Autism	30 (9.5%)
Oppositional defiant disorder	11 (3.5%)
Sleep problems	89 (28.1%)
Concomitant/past psychotherapy for tics, number (%)	148 (46.7%)

Note: Income is reported in Canadian dollars (CAD). *ADHD* attention deficit hyperactivity disorder, *OCD* obsessive-compulsive disorder, *SD* standard deviation.¹ Other medication includes antivirals (amantadine, $n=1$), benzodiazepines ($n=1$), SNRI ($n=1$), muscle relaxant ($n=1$), SARI ($n=3$), and unspecified medications ($n=3$)

Table 3 Average scores on clinical scales

Scale	Mean (SD)
BIS	79.4 (39.1)
PTQ – Total	48.5 (31.5)
PTQ – Motor Tics	28.3 (18.0)
PTQ – Phonic Tics	20.2 (16.6)
RAQ-R	37.2 (15.0)
UPPS-P – Total	57.6 (8.6)
UPPS-P – Negative Urgency	13.0 (2.6)
UPPS-P – Lack of Premeditation	11.5 (2.7)
UPPS-P – Lack of Perseverance	10.6 (3.0)
UPPS-P – Sensation Seeking	9.9 (3.1)
UPPS-P – Positive Urgency	12.6 (2.4)

Note: *BIS* Behavioral Inflexibility Scale, *PTQ* Parent Tic Questionnaire, *RAQ-R* Rage Attack Questionnaire-Revised, *UPPS-P* UPPS-P (Negative Urgency, Lack of Premeditation, Lack of Perseverance, Sensation Seeking, and Positive Urgency) inventory of impulsivity

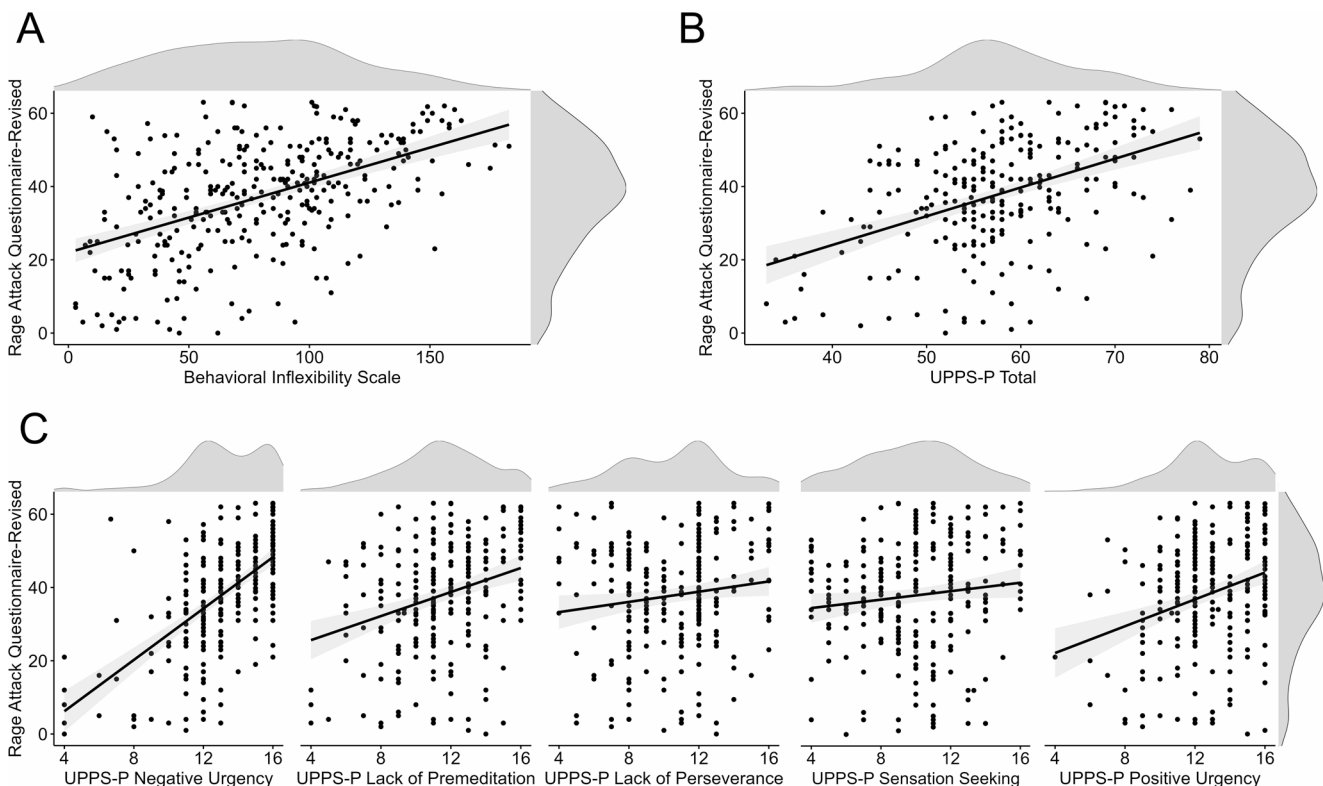


Fig. 1 Associations between explosive outbursts, behavioral inflexibility, and impulsivity. Analyses revealed significant associations between RAQ-R scores and both **(A)** behavioral inflexibility and **(B)** impulsivity, as respectively assessed with the Behavioral Inflexibility Scale and the UPPS-P impulsivity scale. **(C)** Follow-up analyses

revealed significant associations between RAQ-R scores and the four UPPS-P subscales: Negative Urgency, Positive Urgency, Lack of Premeditation, and Lack of Perseverance. There was however no significant association between the RAQ-R and the UPPS-P Sensation Seeking subscale

scores were found in children with co-occurring autism [$t(315)=4.15$, $p<0.001$, $d=0.80$], OCD [$t(315)=3.69$, $p<0.001$, $d=0.46$], sleep problems [$t(315)=3.51$, $p<0.001$, $d=0.44$], internalizing disorders [$t(315)=3.37$, $p<0.001$, $d=0.39$], or ADHD [$t(315)=2.00$, $p=0.047$, $d=0.24$].

For our second aim, valid RAQ-R and UPPS-P data were available from 249 parents. We found a significant association between the RAQ-R and total UPPS-P score [$r(247)=0.45$, $p<0.001$] (Fig. 1B). Exploratory analyses were performed to test the associations between UPPS-P scores and co-occurring (Table S1). Enhanced UPPS-P scores were found in children with co-occurring ADHD [$t(247)=6.71$, $p<0.001$, $d=0.91$], learning disorders [$t(245)=3.35$, $p<0.001$, $d=0.50$], and sleep problems [$t(247)=2.00$, $p=0.044$, $d=0.27$]. When looking at individual UPPS-P subscales, we found significant associations between the RAQ-R and the negative urgency [$r(247)=0.60$, $p<0.001$], positive urgency [$r(247)=0.29$, $p<0.001$], lack of premeditation [$r(247)=0.30$, $p<0.001$], and lack of persistence [$r(247)=0.14$, $p<0.001$] subscales. Only the sensation seeking subscale was not significantly associated with the RAQ-R [$r(247)=0.12$, $p=0.06$] (Fig. 1C).

Table 4 Multiple linear regression with the 5 UPPS-P subscales predicting RAQ-R scores

Predictor	B	SE B	β	t	p
Intercept	-4.34	4.52	—	-0.96	0.34
Negative Urgency	2.88	0.31	0.52	9.24	<0.001
Positive Urgency	-0.28	0.35	-0.05	-0.78	0.43
Lack of Premeditation	0.58	0.30	0.11	1.92	0.06
Lack of Perseverance	-0.04	0.26	-0.01	-0.16	0.87
Sensation Seeking	0.27	0.23	0.06	1.18	0.24

While 4 of the 5 UPPS-P subscales were significantly associated with the RAQ-R, a multiple regression analysis including these 5 subscales revealed that negative urgency predicted explosive outbursts over and above the other dimensions (Table 4). Regarding our third objective, moderation analysis revealed that UPPS-P total scores did not moderate the association between the BIS and the RAQ-R [$F(3,245)=0.007$, $p=0.933$]. Significant effects of BIS on RAQ-R were found at low (-1SD; $B=0.151$, $p<0.001$), mean ($B=0.152$, $p<0.001$), and high (+1SD, $B=0.154$, $p<0.001$) levels of impulsivity, as assessed with the UPPS-P (Fig. 2). Exploratory analyses were conducted with

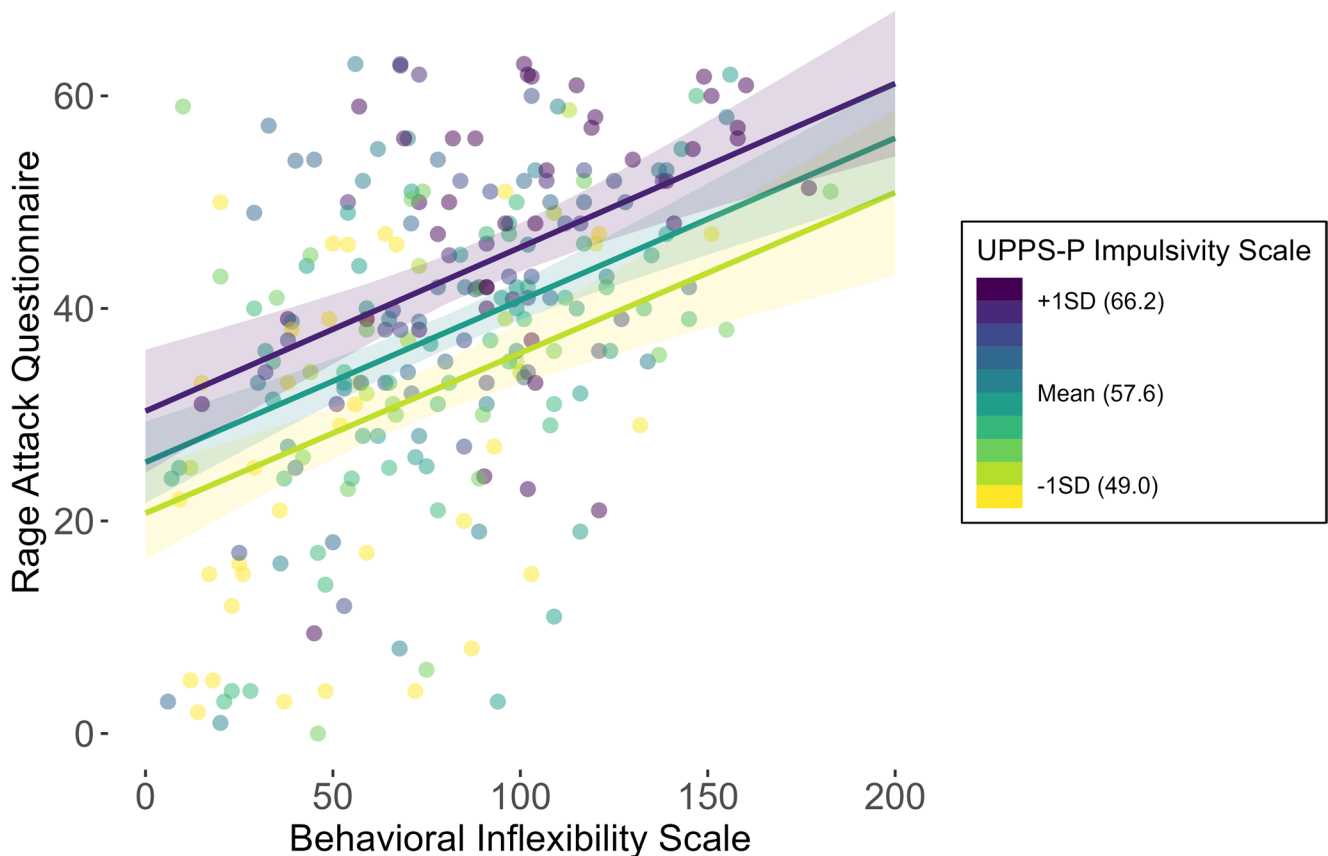


Fig. 2 Testing the moderating effect of impulsivity on the association between behavioral inflexibility and explosive outbursts. Analyses revealed no significant behavioral inflexibility by impulsivity inter-

action. Behavioral inflexibility predicted explosive outbursts at low, average, and high levels of impulsivity

individual UPPS-P scales, but none moderate the association between the BIS and the RAQ-R (see Supplement).

Exploratory analyses revealed small but significant associations between RAQ-R scores and motor [$r(315)=0.12$, $p=0.04$], phonic [$r(315)=0.22$, $p<0.001$], and total [$r(315)=0.18$, $p=0.001$] tic severity (Fig. 3A, B, C). Additionally, the number of co-occurring diagnoses was significantly associated with the RAQ-R [$r(315)=0.26$, $p<0.001$] (Fig. 3D). In terms of individual co-occurring diagnoses, only internalizing disorders [$t(315)=3.88$, $p<0.001$, $d=0.45$], learning disorders [$t(315)=2.52$, $p=0.012$, $d=0.33$] and sleep problems [$t(315)=3.60$, $p<0.001$, $d=0.45$] were significantly associated with increased RAQ-R scores. There was no significant impact of co-occurring ADHD [$t(315)=1.45$, $p=0.15$, $d=0.18$], OCD [$t(315)=0.31$, $p=0.76$, $d=0.04$], autism [$t(315)=0.82$, $p=0.41$, $d=0.16$], developmental coordination disorder [$t(315)=1.60$, $p=0.11$, $d=0.27$], and oppositional defiant disorder [$t(315)=1.05$, $p=0.30$, $d=0.32$] on RAQ-R scores (Fig. 3E). We also assessed whether the co-occurrence of both ADHD and OCD was associated with increased RAQ-R scores but found no significant difference between

children with ADHD+OCD and children with neither co-occurring diagnosis [$t(145)=1.56$, $p=0.25$, $d=0.19$] (Fig. 3F). Also, children currently taking medication were found to have larger RAQ-R scores than unmedicated children [$t(312)=2.83$, $p=0.005$, $d=0.33$] (Fig. 3G).

In terms of demographics, analyses revealed no sex differences on RAQ-R scores [$t(315)=0.66$, $p=0.51$, $d=0.09$] (Fig. 3H). There was also no association between RAQ-R scores and children's age [$r(315)=-0.06$, $p=0.29$] (Fig. 3I). Other exploratory analyses conducted on demographic characteristics are provided in the Supplement.

Discussion

Explosive outbursts are a common feature of TS as well as a leading source of impairment. However, those outbursts have been scarcely investigated, and more research is needed to identify factors that predispose individuals with TS to have explosive outbursts. In the current study, we aimed to assess whether specific individual traits, namely behavioral inflexibility and impulsivity, were associated with an increased

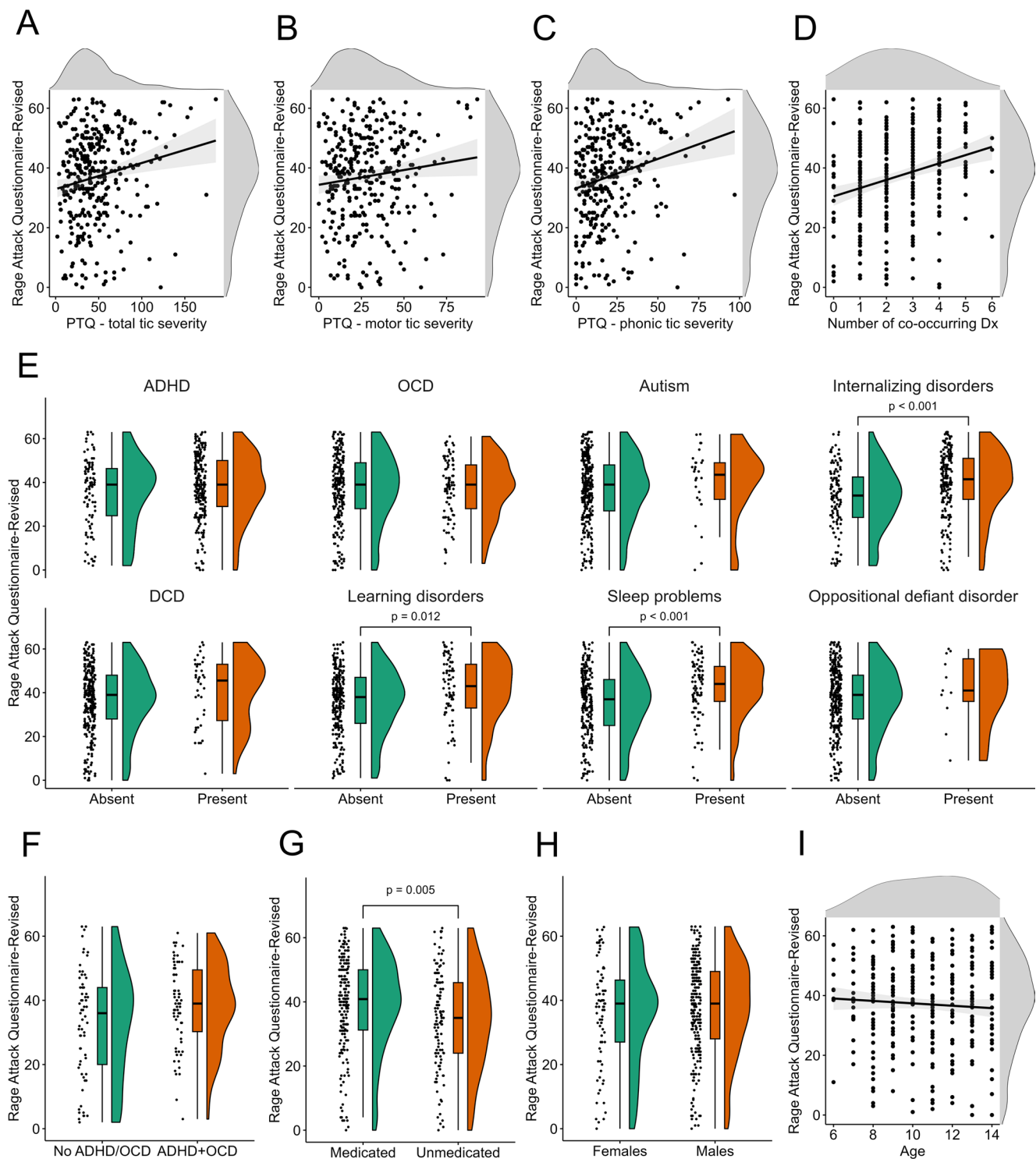


Fig. 3 Associations between RAQ-R scores and clinical and demographic characteristics. There were significant correlations between the RAQ-R and (A) total, (B) motor, and (C) phonic tic severity. (D) The total number of co-occurring diagnoses was significantly associated with higher RAQ-R scores. (E) in terms of individual co-occurring conditions, only internalizing disorders, learning disorders, and

sleep problems were associated with higher RAQ-R scores. (F) Having both co-occurring ADHD and OCD was not found to lead to higher RAQ-R scores relative to having neither co-occurring diagnosis. (G) Medicated children had higher RAQ-R scores than those who were unmedicated. (H) There was no sex difference in terms of RAQ-R scores. (I) Age was no significantly associated with RAQ-R scores

risk of presenting explosive outbursts. We also assessed whether these two traits interacted together and resulted in a more intense presentation of explosive outbursts. Finally, we conducted a series of exploratory analyses to investigate whether several clinical and socio-demographic variables were associated with explosive outbursts.

Early work describing the phenomenology of explosive outbursts highlighted that situations such as not getting one's way or coping with unexpected changes were common precipitants [12]. Building on this prior knowledge, the current study operationalized these observations using the construct of behavioral inflexibility and demonstrated, using a validated measure, that behavioral inflexibility significantly predicted explosive outbursts in children with TS. This pattern suggests that children with a predisposition toward rigid behavioral patterns may be more vulnerable to these episodes. These findings suggest that a predisposition toward rigid behavioral patterns increases vulnerability to explosive episodes, potentially by influencing children's emotional state and ability to cope when facing triggering events. Prior research has highlighted the clinical relevance of irritability in TS, which may act as a precipitant for explosive outbursts [20]. In addition, our results are also consistent with findings in other developmental conditions, in which behavioral inflexibility or insistence on sameness has been associated to heightened irritability and increased likelihood of outbursts [34, 42, 57]. Combined with these earlier findings, our results suggest that behavioral inflexibility may constitute a transdiagnostic risk factor and that it may impact how individuals respond to situational constraints, often leading to emotional dysregulation.

We also found that impulsivity was significantly associated with explosive outbursts in children with TS. This is not surprising as similar findings have been reported among adults [46]. In the current study, we found significant univariate associations between explosive outbursts and all dimensions of the UPPS-P except sensation seeking. Yet, a multiple regression analysis revealed that negative urgency predicted explosive outbursts over and above the other dimensions of impulsivity. This is consistent with what has been reported among adults. While Müller-Vahl, et al. [46] reported significant associations between the RAQ-R and both the negative urgency and lack of perseverance subscales, the former association showed a larger effect than the latter. This result suggests that an inclination to react impulsively when facing negative emotions is an important predictor of explosive outbursts [11]. Similar findings pertaining to negative urgency have been reported in other conditions involving frequent anger outbursts. For instance, intermittent explosive disorder is a condition involving outbursts of physical or verbal aggression that are out of proportion with the triggering event [18]. The

clinical phenomenology of those outbursts is very similar to that of explosive outbursts among children with TS [19]. In individuals with intermittent explosive disorder, negative urgency was found to be higher than in unaffected controls and associated with poor anger control [51]. These findings suggest that impulsivity, and especially negative urgency, is an additional transdiagnostic factor.

While we found significant associations between explosive outbursts and both behavioral inflexibility and impulsivity, we did not find impulsivity to moderate the association between behavioral inflexibility and explosive outbursts. Behavioral inflexibility thus appears to contribute to explosive outbursts independently of impulsivity. One potential explanation for this null effect is the relatively high average UPPS-P score in our sample as well as the limited variance in scores. There might not have been enough differentiation between participants with low and high impulsivity to probe a moderating effect of impulsivity on the association between behavioral inflexibility and explosive outbursts.

Exploratory analyses revealed that several co-occurring conditions were associated with a more severe explosive outbursts presentation. Children with internalizing disorders such as anxiety or depression, learning disorders, and sleep problems were identified with enhanced RAQ-R scores relative to children who did not have these co-occurring diagnoses. Surprisingly, we did not find explosive outbursts to be increased in children with co-occurring ADHD, OCD, or both conditions. This contrasts with previous findings showing associations between anger outbursts or aggression and ADHD [5, 17, 27, 43, 59, 60] or OCD [35, 59]. It is however important to distinguish between explosive outbursts and other forms of aggressive behaviors, such as those that can be seen in conduct disorder or severe oppositional defiant disorder. In children with explosive outbursts, their temper during such episodes is usually disproportionate to the triggering event and reflect a loss of emotional control rather than a deliberate intent to harm someone verbally or physically [29]. This contrasts with disruptive behavior disorders, where aggression is typically more purposeful or instrumental [36]. Previous research has suggested that disruptive behavior disorders may not be an integral part of TS but would be more related to the presence of ADHD as well as a family history of aggressive and violent behavior and forensic encounters [54]. Given that explosive outbursts in TS have often been included under the umbrella of aggressive behaviors, it appears possible that their association with ADHD may have been overestimated and conflated with other types of aggressive behaviors. Additionally, our sample had a prevalence of co-occurring ADHD of almost 70%, which is larger than what is typically observed in community samples [16]. This may be explained, at least partially, by the fact that the province of Quebec, where the research

was conducted and where an important proportion of participants came from, has one of the highest ADHD prevalence worldwide [9, 25, 33]. Furthermore, it is possible that parents of children with TS+ADHD were more likely to perceive a study on explosive outbursts as relevant to their child's difficulties, hence the oversampling of co-occurring ADHD. Other factors may also explain why co-occurring diagnoses of ADHD and/or OCD were not associated with greater severity of explosive outbursts. Some parents may have judged that their child likely meets the diagnostic criteria for ADHD or OCD even without a formal diagnosis. In addition, because this project was advertised as a study focusing on explosive outbursts and their impact on family functioning, the children whose parents chose to participate in our study may present with more severe TS and associated conditions than the average child with TS. Thus, individual characteristics associated with more complex forms of TS may have overshadowed the usual effect of ADHD and OCD on explosive outbursts. Furthermore, we did not collect a measure of ADHD or OCD symptom severity, and we cannot exclude that the severity of symptoms would be associated with explosive outbursts. Nevertheless, we found that impulsivity, which was larger among children with co-occurring ADHD, was significantly associated with RAQ-R scores. Similarly, even if autism and OCD were not associated with larger RAQ-R scores, we found that behavioral inflexibility, which was higher in children with co-occurring autism or OCD, was significantly associated with explosive outbursts. This suggests that common transdiagnostic traits associated with neurodevelopmental and mental health conditions are stronger predictors of explosive outbursts than an actual diagnosis of a co-occurring condition.

Our findings must be interpreted in the light of some limitations. Conducting this study online allowed us to obtain a large sample with efficient use of time and resources, but it comes with the limitations inherent to online research. We could not validate participants' diagnoses and did not obtain clinician-rated measures of tic and associated symptoms severity. Also, all measures collected in this study were parent-report. It would have been interesting to contrast those with self-report measures. Finally, as we mentioned above, the proportion of children with co-occurring ADHD in our study was larger than what is typically observed in community studies of TS. Our sample is therefore less representative of youth with TS who have limited inattentive and hyperactive symptoms.

In conclusion, we identified factors, namely behavioral inflexibility and impulsivity, that predispose to explosive outbursts among youth with TS. Our results also revealed that these transdiagnostic traits may predict explosive outbursts over and above co-occurring diagnoses of neurodevelopmental disorders. Sudden anger outbursts are common

not only in TS but also across other neurodevelopmental conditions, such as autism [47, 62] and ADHD [31, 58]. Similarly, behavioral inflexibility and impulsivity are also common transdiagnostic traits observed across neurodevelopmental conditions [10, 38, 63, 64, 66]. It remains unclear whether the explosive outbursts experienced by an important proportion of individuals with TS can be attributed to characteristics that are proper to TS. Future work should assess the anger phenomena across the spectrum of neurodevelopmental conditions in order to identify similarities or differences in the presentation of anger outbursts between conditions. This may allow understanding whether explosive outbursts in TS are a specific feature driven by specific aspects of the TS symptomatology.

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Data availability The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Competing interests The authors declare no competing interests.

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